Spectral Distribution-Aware Estimation Network for Hyperspectral Anomaly Detection

Weiying Xie, *Member, IEEE*, Shuran Fan, Jiahui Qu, Xianyun Wu, Yanli Lu, and Qian Du, *Fellow, IEEE*

***Abstract*— Recently developed deep learning-based hyperspectral anomaly detection (HAD) methods typically include two steps where the deep feature extraction is not designed specifically for the HAD task. In this article, we propose a spectral distribution-aware estimation network (SDEN) that does not conduct feature extraction and anomaly detection in two separate steps but instead learns both jointly to estimate anomalies directly in an end-to-end manner without postprocessing. The unified framework can ensure that the extracted features serve better for anomaly detection. To preserve the distribution of hyperspectral images (HSIs) during dimensionality reduction, the SDEN introduces a spectral distribution (SD)-aware module imposed with a local-invariant constraint. More specifically, we adopt Markov chain Monte Carlo (MCMC) that enables the SD module to better estimate the distribution of the complex HSIs. Considering the powerful representation capability of Gaussian mixture model (GMM), the SDEN leverages it to establish an estimation module in the deep latent space where the anomaly resides in low density while the background not. We demonstrate that the SDEN yields competitive and highly promising results in comparison with the anomaly detection benchmarks.**

***Index Terms*— Estimation network, hyperspectral anomaly detection (HAD), joint optimization, local-invariant constraint, spectral distribution (SD).**

I. Introduction

With advanced imaging technologies, hyperspectral images (HSIs) can detect hidden and camouflaged targets through fine spectral characteristic profiles [1]–[6]. In the case that the prior target knowledge is unavailable, hyperspectral anomaly detection (HAD) has to be applied [7]–[11]. It is used to find anomalous pixels that are significantly different from the background in spectral signatures. Under the blind environment, there are many challenges in HAD due to the high dimensionality and complexity of HSIs.

To this end, the existing HAD methods intend to ensure detection capability, which can be categorized into either one step or two steps. One-step approaches perform HAD directly based on the input HSIs. Among these methods, the Reed–Xiaoli (RX) detector calculates the mean and variance of samples in the input HSIs under the assumption that the background obeys a Gaussian distribution [12]. The local RX detector (LRX) estimates anomalies in a local window [13]. The collaborative representation detector (CRD) performs HAD with the assumption that an anomaly cannot be represented by its neighborhood pixels, while the background can [14]. These one-step approaches ensure that all operations serve to detect anomalies. However, traditional methods have limited ability to characterize high-dimensional and complex data, thus making it difficult to achieve performance improvements.

Different from the abovementioned one-step approaches, two-step methods extract features of the input HSIs and then generate HAD decision on the extracted features. Zhang *et al.* [15] utilized the low-rank and sparse attributes to extract features and then employed the Mahalanobis distance to detect anomalies. Li *et al.* [16] combined the low-rank background with a mixture noise model to characterize more complex distribution. Xu *et al.* [17] introduced an anomaly detector based on the low-rank and sparse representation with adding a sparsity-inducing regularization term. Recently, Li *et al.* [18] proposed a prior-based tensor approximation (PTA) that decomposed HSIs into a background tensor and an anomaly tensor to perform HAD. The emerging two-step approaches also include attribute and edge-preserving filters-based detector (AED) [19] and an adaptive weighting detector based on the structured tensor and guided filtering algorithm (STGF) [20], both of which first reduce data dimensionality by principal component analysis (PCA) or band selection and then distinguish anomalies through a combination of attribute filtering and differential operation. Huang *et al.* [21] explored spectral, extended morphological profiles (EMPs), and Gabor texture features for subsequent anomaly detection. More recently, deep neural networks (DNNs) have proved to successfully extract features from complex data [22]–[25]. In the HAD task, the nonlinear mapping relationship between the input HSIs and reconstructed ones is first learned by DNNs, and the latent features or the reconstruction residuals

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Weiying Xie, Shuran Fan, Jiahui Qu, and Xianyun Wu are with the State Key Laboratory of Integrated Service Networks, Xidian University, Xi'an 710071, China (e-mail: wyxie@xidian.edu.cn; shuranf@stu.xidian.edu.cn; jhqu@xidian.edu.cn; xywu@mail.xidian.edu.cn).

Yanli Lu is with the College of Air and Missile Defense, Air Force Engineering University, Xi'an 710051, China (e-mail: luyanlihgh@163.com).

Qian Du is with the Department of Electronic and Computer Engineering, Mississippi State University, Starkville, MS 39759 USA (e-mail: du@ece.msstate.edu).

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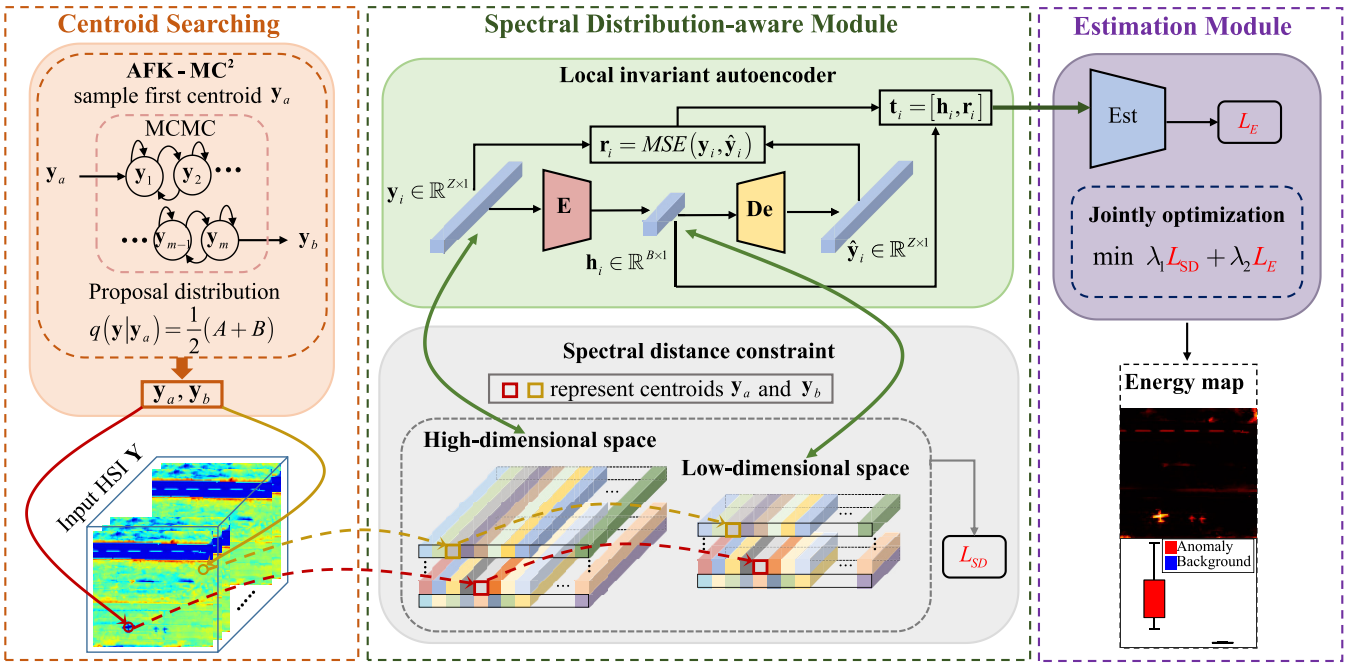


Fig. 1. Workflow of the proposed SDEN method in HSI.

are used to detect anomalies [26]–[31]. Notably, the feature extraction and detection steps are separate. Although these two-step methods alleviate the curse of dimensionality, separated optimization leads to generated features not being fully specific to the anomaly detection task. In addition, the distribution pattern of the input HSIs may be destroyed, while the data are reduced in dimensionality.

To address these issues essentially, we propose a novel spectral distribution-aware estimation network (SDEN) for HAD in an end-to-end manner. Unlike previous methods where feature extraction and anomaly estimation are independent, the proposed SDEN optimizes both jointly, that is, SDEN directly outputs its anomaly probability estimate for each input spectral vector. Fig. 1 shows the workflow of the proposed SDEN. In SDEN, spectral local invariance is imposed in the low-dimensional manifold to preserve the distribution pattern of the input HSIs. More importantly, we introduce a spectral distribution (SD)-aware module based on the assumption that the centroids in the high-dimensional HSIs are still the centroids in the low-dimensional manifold. Concretely, the spectral distance between each spectral vector and the centroid vector remains constant in the low-dimensional manifold. In particular, considering the complexity of HSIs, we exploit Markov chain Monte Carlo (MCMC) to model the SD in the input HSIs. Meanwhile, for accurately distinguishing anomaly and background, we induce a seeding approach called assumption-free K $-$ MC^2 (AFK $-$ MC^2) to find better centroids. Besides, a learnable estimation module is incorporated to produce a probability density for each spectral vector to be an anomaly in an end-to-end manner. Finally, SDEN is updated by stochastic gradient descent (SGD) [32]. Compared with the state-of-the-art methods, the proposed anomaly detector achieves superior performance.

The contributions of this article are threefold.

1) We propose a novel HAD network, called SDEN, which imposes the SD pattern of the input HSIs for joint optimization of feature extractor and anomaly detector.
2) A universal SD-aware module is introduced and combined with local-invariant constraint to maintain the data structure of the input HSIs during dimensionality reduction.
3) The learnable estimation module is established considering the complexity of HSIs, ultimately leading to superior HAD performance.

The remainder of this article is organized as follows. Section II gives a brief description of MCMC and AFK$-$MC^2. The proposed SDEN is introduced in Section III. Section IV presents the experimental results and comprehensive analyses. Section V gives several conclusions.

II. Related Work

In this section, we briefly describe the techniques employed in this article. This includes MCMC and assumption-free K $-$ MC^2 (AFK $-$ MC^2).

A. Markov Chain Monte Carlo

MCMC is a random sampling method that has a wide range of applications in deep learning and aims to find samples obeying the given probability distribution [33]. Based on the fact that the Markov chain can converge to a stationary distribution under certain conditions, MCMC constructs a Markov chain with a transition matrix $\mathbf{P}$ whose stationary distribution is the target distribution $q(x)$. In this case, we can start from an initial state x_0 and move along the Markov chain to generate a transition sequence. Consequently, this transition sequence is considered as the samples obeying the target distribution $q(x)$. Since the transition matrix $\mathbf{P}$ (where $p(i, j)$ represents the probability of transition from state i to state j) and the target distribution $q(x)$ may not satisfy the

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detailed and stable condition of Markov chain, an acceptance rate $\alpha(i, j)$ is employed to induce the satisfaction of condition and the formulation is established as

$$q(i)p(i, j)\alpha(i, j) = q(j)p(j, i)\alpha(j, i). \tag{1}$$

However, considering the issue that the acceptance rate $\alpha(i, j)$ of the Markov chain in the transfer process may be too small and it will cause the Markov chain to take a long time to converge to the steady-state distribution, MCMC enlarges the acceptance rate $\alpha(i, j)$ and $\alpha(j, i)$ in the same proportion. In other words, the optimal situation is to increase the larger one of the two numbers to 1. Then, the acceptance rate can be defined as

$$\alpha(i, j) = \min\left\{\frac{p(j)q(j, i)}{p(i)q(i, j)}, 1\right\}. \tag{2}$$

According to the definition of the acceptance rate $\alpha(i, j)$, MCMC generates a Markov chain with transition matrix $\mathbf{P}$ whose stationary distribution is exactly the target distribution $q(x)$. Due to the efficiency and applicability for sampling of multivariate random variables, MCMC is adopted in our model to better and more quickly represent the complex and high-dimensional HSIs.

B. Assumption-Free $K - MC^2$

Due to simplicity, the seeding step of k-Means [34] and k-means++ [35] is widely used. Based on the Euclidean distance, k-means updates the centroids with random initials. However, considering the fact that it only can find a local optimum instead of a globally optimal solution, k-means++ adopts a sampling method called D^2-sampling whose core is the distance between centroids should be as far as possible. These algorithms are time-consuming in selecting centroids. As a representative improved version, the MCMC ($\mathrm{K-MC}^2$) approximation [36] reduces the computational time of seeding. However, since $\mathrm{K-MC}^2$ adopts a uniform proposal distribution, this algorithm exists a potential disadvantage that those centroids with a small number of samples may not be selected as candidate nodes. In this case, with adding a distance-related distribution to the original uniform proposal distribution, assumption-free $\mathrm{K-MC}^2$ ($\mathrm{AFK-MC}^2$) [37] can improve the seeding robustness and adapt to different data distribution assumptions, which is applied in our model to estimate centroids.

III. Proposed Method

Since two-step approaches in existence fail to dig out the goal-directed characterization embedded in the low-dimensional representation space causing suboptimal solution, we construct SDEN to jointly optimize dimensionality reduction and anomaly detection to ensure each component benefits the other. As shown in Fig. 1, the proposed SDEN is composed of three major components: centroid searching, SD-aware module, and estimation (E) module.

Let an original hyperspectral dataset $\mathbf{Y} \in \mathbb{R}^{M\times N\times Z}$ be denoted as $\mathbf{Y} = \left[\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_K\right]$, where $\mathbf{y}_i \in \mathbb{R}^{Z\times 1}$ $\left(i = 1, 2, \ldots, K, K = M \times N\right)$ represents the ith spectral vector of the HSI. We formally describe our model as

$$\begin{aligned} &\min\ \lambda_1 L_{\mathrm{SD}} + \lambda_2 L_E \\ &\text{s.t. } C_{\mathrm{LI}} < \varepsilon \end{aligned} \tag{3}$$

where the first component represents the spectral distance constraint (SDC) to maintain the data structure during dimensionality reduction, and the second component characterizes the probability density and serves as an estimation module for HAD. The constraint takes advantage of the local invariance as a premise to keep the corresponding spectral connectivity. We optimize C_{LI} until it is smaller than ε, which is a sufficiently small value. We introduce more details of the SDEN framework in the following.

A. Centroid Searching

The centroid searching process is prepared for SD module and intends to obtain the centroids of the anomaly and background by a distribution-based seeding method. In order to improve the accuracy of seeding and reduce computational complexity, we employ a fast and accurate seeding algorithm called $\mathrm{AFK-MC}^2$ to decide the centroids. The principle of seeding is based on an observation that the further away from the previous centroid, the more likely a pixel will be selected as the next centroid. As the key of $\mathrm{AFK-MC}^2$, D^2-sampling focuses on the selection of centroids. However, since the universal D^2-sampling is originally a computationally intensive task, a novel sampling method based on MCMC is applied to approximate the process of D^2-sampling.

Fig. 2 shows the procedure of seeding according to MCMC. Aiming to generate samples that obey the given distribution, MCMC is adopted to obtain two centroids corresponding to anomaly and background obeying the target distribution $p(\mathbf{y})$. Notably, the initial state $\mathbf{y}_1$ selected at random is determined as the first centroid $\mathbf{y}_a$. In the process of state transition, if the state satisfies the transfer conditions, the next state $\mathbf{y}_2$ will equal $\mathbf{x}$, which is sampled using the proposal distribution $q\left(\mathbf{x}\middle|\mathbf{y}_1\right)$. Subsequently, according to the Markov chain of length M sampled by the loop of previous steps, we obtain the second centroid $\mathbf{y}_b$ equaling $\mathbf{y}_m$, which represents the last state. In particular, for improving the robustness of clustering and adapting to different data distribution, a distance-related distribution is added to the uniform distribution. For a spectral vector $\mathbf{y} \in \mathbf{Y}$, the proposal distribution is defined as

$$q(\mathbf{y}|\mathbf{y}_a) = \frac{1}{2}(A + B) \tag{4}$$

$$A = \frac{d(\mathbf{y}, \mathbf{y}_a)^2}{\sum_{\mathbf{y}'\in\mathbf{Y}} d(\mathbf{y}', \mathbf{y}_a)^2} \tag{5}$$

$$B = \frac{1}{|\mathbf{Y}|} \tag{6}$$

where $\mathbf{y}_a \in \mathbf{Y}$ is the first centroid sampled uniformly at random. Term A, representing for the true D^2-sampling distribution with regard to the first centroid $\mathbf{y}_a$, guarantees that any dataset can begin with the best possible proposal distribution for later iterations. Term B is designed to regularize the proposal distribution. Consequently, we obtain two centroids

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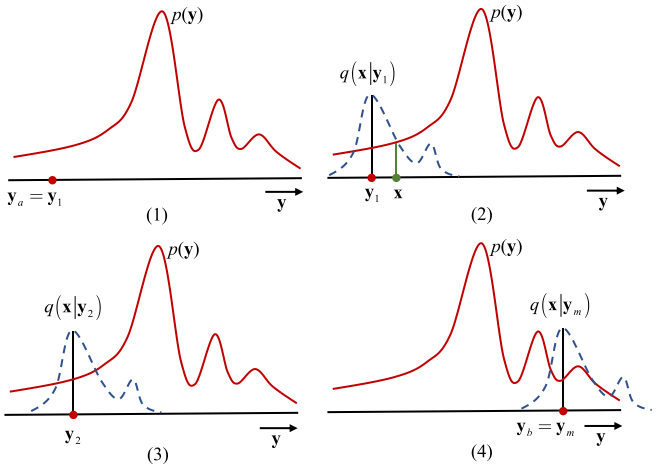


Fig. 2. Procedure of seeding according to MCMC: (1) consider the initial state $\mathbf{y}_1$ as the first centroid $\mathbf{y}_a$, (2) sample $\mathbf{x}$ using the proposal distribution $q\left(\mathbf{x}\middle|\mathbf{y}_1\right)$, (3) equal $\mathbf{y}_2$ to $\mathbf{x}$ if the state satisfies the transfer condition, and (4) determine the last state $\mathbf{y}_m$ as the second centroid $\mathbf{y}_b$.

$\mathbf{y}_a$ and $\mathbf{y}_b$, which can be determined as centroids of anomaly and background in the original hyperspectral space.

B. SD-Aware Module

The SD module is constructed to extract features of the input hyperspectral data and retain the original data distribution. Based on the fact that the low-dimensional representation learned from the original autoencoder (AE) has a proportion of probability to fail to capture the intrinsic structure of the input data, a local-invariant AE (LIAE) is applied to preserve the local connectivity. In addition, the corresponding relationship constraint of centroids is added to keep the data distribution and improve the detection performance. The SD module consists of the LIAE and the SDC.

1) Local-Invariant Autoencoder: Under the assumption that anomalies are distinct spectrally from the surrounding background and the occurrence of anomalies is in a low probability, we design this LIAE to shorten the distance between spectra with similar signatures and push away those with different signatures. The LIAE is composed of an encoder, a decoder, and a regularizer to capture the local structure information and preserve the local connectivity.

The AE is chosen to learn a low-dimensional representation, which often gives a better representation of the input than the original input $\mathbf{Y}$. Both the encoder ($\mathbf{E}$) and the decoder ($\mathbf{De}$) deploy multilayer fully connected neural networks followed by the tanh activation. The encoding and decoding process can be formulated as

$$\begin{aligned}\mathbf{H} &= f(\mathbf{W}\mathbf{Y}+b)\\ \hat{\mathbf{Y}} &= f\left(\hat{\mathbf{W}}\mathbf{H}+\hat{b}\right)\end{aligned} \tag{7}$$

where the encoder ($\mathbf{E}$) maps input $\mathbf{Y}$ in the original high-dimensional space to $\mathbf{H}$ in the latent low-dimensional space with the weight matrix $\mathbf{W}$ and the bias term $\boldsymbol{b}$. Then, the decoder ($\mathbf{De}$) maps $\mathbf{H}$ back to the original input layer $\hat{\mathbf{Y}}$ with the weight matrix $\hat{\mathbf{W}}$ and the bias term $\hat{\boldsymbol{b}}$.

However, the original AE often fails to reveal the local structural information with a universal fidelity loss function, which results in poor performance of AE on HAD. Based on the manifold assumption, data samples are considered locating on the neighborhood of a low-dimensional manifold in the high-dimensional space. Since the representation should discover and preserve the latent geometrical structure of the original data, the local invariance intents to capture the intrinsic low-dimensional manifold embedded in the high-dimensional space. Especially, the local invariance can be summarized as a reasonable thought that if $\mathbf{y}_i$ and $\mathbf{y}_j$ are close to each other in the intrinsic structure of data distribution, their corresponding representation in the latent space $\mathbf{h}_i$ and $\mathbf{h}_j$ is also close to each other. In this case, we impose a local-invariant constraint on the loss function of the original AE, and C_{LI} in (3) is described as the following formulation:

$$C_{\mathrm{LI}} = \frac{1}{K}\sum_{i=1}^{K}\|\mathbf{y}_i-\hat{\mathbf{y}}_i\|_2 + \frac{\alpha}{K}\sum_{i=1}^{K}\sum_{j=1}^{K}\mathbf{g}_{ij}\left\|\mathbf{h}_i-\mathbf{h}_j\right\|_2 \tag{8}$$

where $\hat{\mathbf{y}}_i$ means the reconstruction of $\mathbf{y}_i$, K indicates the number of input data, α is regarded as the weight coefficient of constraint term and is set to 3, and $\mathbf{g}_{ij}$ is the (i, j)th entry of an affinity matrix $\mathbf{G} = (\mathbf{g}_{ij}) \in \mathbb{R}^{K\times K}$ and represents the feature similarity between input samples $\mathbf{y}_i$ and $\mathbf{y}_j$. Specifically, if $\mathbf{y}_j$ lies in $\mathbf{y}_i$'s δ-nearest neighbor set, which contains all the spectra whose distances to $\mathbf{y}_i$ are less than δ, i.e., $\|\mathbf{y}_i-\mathbf{y}_j\| < \delta$, $\mathbf{g}_{ij} = 1$. Otherwise, $\mathbf{g}_{ij} = 0$. Besides, $\mathbf{G}$ is a sparse weight matrix, which only works on few nearest neighbors of each input sample, and it plays an essential role in capturing the original relationship from the hyperspectral space. Since this formulation aims at maintaining the corresponding connectivity based on local invariance, we employ it as a precondition to the constraint of our model.

Considering the issue that anomalies after reconstruction may produce high reconstruction errors, we apply the reconstruction error $\mathbf{r}_i$ to enlarge the discrimination between anomalies and background, and it can be formed with mean squared error (MSE) as

$$\mathbf{r}_i = \mathrm{MSE}(\mathbf{y}_i, \hat{\mathbf{y}}_i). \tag{9}$$

To take full advantage of the complementary information embedded in the local-invariant low-dimensional representations, the latent representation $\mathbf{h}_i$ and the reconstruction error $\mathbf{r}_i$ given above are concatenated to generate $\mathbf{t}_i$

$$\mathbf{t}_i = [\mathbf{h}_i, \mathbf{r}_i]. \tag{10}$$

Subsequently, $\mathbf{t}_i$ is fed into the following estimation network.

2) Spectral Distance Constraint: Considering the fact that the original distribution pattern may be destroyed in the process of dimensionality reduction, we propose an SDC to maintain the data distribution. Specifically, a reasonable assumption is that if an input spectral vector is confirmed as the centroid in the high-dimensional space, it is also the centroid in the mapped low-dimensional space. Since we have already obtained the two centroids $\mathbf{y}_a$ and $\mathbf{y}_b$ that represent the centroids of anomaly set and background set in the input

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space, we can draw the conclusion that $\mathbf{h}_a$ and $\mathbf{h}_b$, denoted as the latent representation of $\mathbf{y}_a$ and $\mathbf{y}_b$, are regarded as the two centroids in the hidden space. Meanwhile, by calculating the included angle between two spectral vectors, spectral angle mapper (SAM) intends to measure the spectral similarity between them. The smaller the angle, the more similar the two spectra are. In this case, SAM is employed to keep the correspondence of distance between centroids and any other spectral vector in high-dimensional space and low-dimensional space. The SDC is formed as

$$S_\beta = \frac{1}{K-2}\sum_{i=1;i\neq\beta}^{K} \arccos \frac{\mathbf{y}_\beta^{\mathrm{T}}\mathbf{y}_i}{\sqrt{(\mathbf{y}_\beta)^{\mathrm{T}}\mathbf{y}_\beta}\sqrt{(\mathbf{y}_i)^{\mathrm{T}}\mathbf{y}_i}} \tag{11}$$

$$\hat{S}_\beta = \frac{1}{K-2}\sum_{i=1;i\neq\beta}^{K} \arccos \frac{\mathbf{h}_\beta^{\mathrm{T}}\mathbf{h}_j}{\sqrt{(\mathbf{h}_\beta)^{\mathrm{T}}\mathbf{h}_\beta}\sqrt{(\mathbf{h}_j)^{\mathrm{T}}\mathbf{h}_j}} \tag{12}$$

where $\beta = a, b$, S_β refers to SAM distance in the input space, and $\hat{S}_\beta$ refers to SAM distance in the latent space. Consequently, the first component in (3) can be established as

$$\lambda_1 L_{\mathrm{SD}} = \lambda_1 |\mathrm{SAM1} - \mathrm{SAM2}| \tag{13}$$

$$\mathrm{SAM1} = \frac{1}{2}\sum S_\beta \tag{14}$$

$$\mathrm{SAM2} = \frac{1}{2}\sum \hat{S}_\beta \tag{15}$$

where λ_1 represents the hyperparameter and is set to 3 in our experiments. SAM1 means the average SAM distance of anomalies and background in the original space and SAM2 means the average SAM distance of anomalies and background in the latent space. L_{SD} controls the distance of the spectral vector with similar spectral signatures and aims to improve the accuracy of anomaly detection.

C. Estimation (E) Module

Since the distribution of hyperspectral data is complex and may not be multivariate Gaussian, we build an estimation (E) module with Gaussian mixture model (GMM) due to its powerful representational capability. Depending on $\mathbf{t}_i$ fed from LIAE and the number of mixture components l, we apply a linear softmax function to the last layer of the neural network NN to make estimation

$$\xi_i = \mathrm{Est}(\mathbf{t}_i) = \mathrm{softmax}(NN(\mathbf{t}_i)) \tag{16}$$

where $\xi_i \in \mathbb{R}^{l\times 1}$ means an l-dimensional vector for the soft mixture component membership of the spectral vector $\mathbf{y}_i$. With a batch of k spectra and their membership prediction, the necessary parameters of estimation network can be defined as

$$N_l = \sum_{i=1}^{k} \xi_{il} \tag{17}$$

$$\psi_l = \frac{N_l}{k} \tag{18}$$

$$\boldsymbol{\mu}_l = \frac{1}{N_l}\sum_{i=1}^{k} \xi_{il}\mathbf{t}_i \tag{19}$$

$$\Sigma_l = \frac{1}{N_l}\sum_{i=1}^{k} \xi_{il}(\mathbf{t}_i - \boldsymbol{\mu}_l)(\mathbf{t}_i - \boldsymbol{\mu}_l)^{\mathrm{T}} \tag{20}$$

where ψ_l, $\boldsymbol{\mu}_l$, and Σ_l are mixture probability, mixture mean vector, and mixture covariance matrix for component l ($l = 1, 2, \ldots, L$), respectively. Since the probability of anomaly occurrence is low and the probability of background occurrence is high, it is easy to identify spectra with low probability density as anomalies. Thus, we obtain the anomaly score of each $\mathbf{t}_i$ by calculating the negative log likelihood of probability density

$$E(\mathbf{t}_i) = -\log\left(\sum_{l=1}^{L} \frac{\psi_l}{\sqrt{|2\pi\,\Sigma_l|}} e^{\left(-\frac{1}{2}(\mathbf{t}_i-\boldsymbol{\mu}_l)^{\mathrm{T}}\Sigma_l^{-1}(\mathbf{t}_i-\boldsymbol{\mu}_l)\right)}\right). \tag{21}$$

Based on the fact that spectra with low density are more likely to be anomalies and spectra with high density are more likely to be background, we can obtain an energy map with scores of anomalous pixels according to every value of $E(\mathbf{t}_i)$. Therefore, the second term of our model can be detailed as

$$\lambda_2 L_E = \lambda_2 \sum_{i=1}^{K} E(\mathbf{t}_i) \tag{22}$$

where λ_2 means the hyperparameter and is set to 0.1. For the sake of solving the local optimal problem, we jointly optimize the SD module and the E module with the constraint of local invariance. During the training procedure, the SD module can maintain the data distribution and extract features of the latent space. Meanwhile, the E module can perform anomaly detection based on the probability density derived from the $\mathbf{t}_i$, which is fed from the SD module. In the nutshell, the E module and the SD module combined with local invariance mutually serve each other, which leads to a superb detection performance of SDEN.

IV. Experiments

In this section, we conduct abundant experiments on the proposed SDEN and the most advanced or representative algorithms in existence to compare and evaluate the anomaly detection performance from qualitative and quantitative perspectives.

A. Datasets Preparation

Four real HSIs are adopted in our experiments with noisy bands being removed and reference maps of the original data being manually labeled. More specific details are listed as follows.

1) Gulfport Dataset: The first dataset was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor from Gulfport, Southern Mississippi, USA, in 2010. In this image, the total area is of 100 × 100 pixels and we take full use of 191 spectral bands with wavelengths varying from 400 to 2500 nm. Besides, the spatial resolution is 3.4 m/pixel. The anomalous targets, including three airplanes in different scales, account for 60 pixels in total.

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2) Texas Coast-2 Dataset: The second dataset was recorded from the Texas Coast, USA, in 2010 by the AVIRIS sensor. With the noisy bands having been removed, this image has a spatial size of 100 × 100 and spectral bands of 207. Moreover, the ground resolution is 17.2 m/pixel.

3) Gainesville Dataset: The third dataset was collected by the AVIRIS sensor from Gainesville, FL, USA, in 2010. After removing the noisy bands, this dataset, including 191 spectral channels, has a spatial size of 100 × 100 with a ground resolution of 3.5 m.

4) HYDICE Dataset: The fourth dataset obtained by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor shows the overview of an urban area, CA, USA. With a total of 162 spectral channels in spectral coverage varying from 400 to 2500 nm, the image consists of 80 × 100 pixels that preserve all useful information of the original data, including vegetation, roadways, and several vehicles. Among the pixels, vehicles and roofs are regarded as anomalous targets, which account for 0.2375%.

B. Experimental Setup

1) Comparative Methods: Six state-of-the-art HAD approaches are applied for comparison, i.e., RX [12], low-rank and sparse matrix decomposition-based Mahalanobis distance for hyperspectral anomaly detection (LSMAD) [15], low-rank and sparse representation-based anomaly detector (LRASR) [17], low-rank and sparse decomposition model with mixture of Gaussian (LSDM-MoG) [16], PTA [18], and adversarial AE (AAE) [29]. As a basic algorithm for anomaly detection, the RX can have a good detection on simple scenes by estimating the global background statistical distribution. With solving the anomaly contamination and the inverse covariance matrix problem, the LSMAD extracts the knowledge from both background and anomalies to perform HAD. The parameter setting is consistent with [15]. As for the LRASR algorithm, parameters K and P play a crucial role in the dictionary step and they are set to 15 and 20. The regularization parameters β and λ are both fixed as 0.1. For the LSDM-MoG method, the initial rank l_0 is empirically set to 60 and the initial number of mixture Gaussian noise K is fixed as 4. For the PTA method, the hyperparameters α and τ are set to 1 and $\beta = 0.01$. Meanwhile, the optimal values of μ are different for each dataset and still need to be tuned individually. Moreover, the main parameters of AAE method are selected according to [29].

2) Performance Evaluation Criterion: In this section, we employ the receiver operating characteristic (ROC) [38] curve, the area under ROC curve (AUC) [39] and the box plot as our evaluation metrics to quantitatively and qualitatively investigate the anomaly detection performance of the proposed method and its comparison algorithms. As the most commonly used evaluation index, ROC curve takes true positive rate (TPR) as the horizontal axis and false positive rate (FPR) as the vertical axis, where TPR represents the probability of detection and FPR represents the probability of false alarm. With an appropriate threshold (τ), the ROC curve can intuitively reflect the accuracy of various detectors through the diagram. By computing the whole area under the ROC curve, AUC acts as a performance indicator that measures the pros and cons of the detectors. Notably, the AUC of (TPR, FPR) is expected to approach to 1 and the AUC of (FPR, τ) is expected to approach to 0. Since box plot [40] is mainly used to reflect the characteristics of the original data distribution, we apply it to measure the ability to separate anomaly and background of detectors.

C. Parameter Analysis

In this section, we focus on the sensitivity of crucial parameters and analyze their effects on the experimental results of the proposed method. These parameters include the number of hidden nodes in the process of dimensionality reduction, the number of hyperparameter λ_1 in the SD module, and the number of hyperparameter λ_2 in the E module.

1) Number of Hidden Nodes: As a pivotal parameter in the proposed method, the number of hidden nodes impacts greatly on the learning ability of the feature extractor LIAE. With a small value, the LIAE may fail to capture a discriminative representation in the original input space. However, with a large value, the LIAE may not effectively suppress the redundant background information. Hence, we set the number of hidden nodes to 18, 19, 20, 21, and 22 and evaluate the AUC score of (TPR, FPR) on each experimented dataset. As shown in Fig. 3(a), all the datasets obtain the best detection result when the number of hidden nodes reaches 20.

2) Number of Gaussian Components l: The number of Gaussian mixture components l plays a significant role in improving the ability of E module to describe hyperspectral data distribution. With an appropriate value, E module can effectively distinguish the anomaly and background. Therefore, we set the value to 3, 5, 7, 9, 11, and 13 and evaluate the AUC score of (TPR, FPR) on each experimented dataset. As shown in Fig. 3(b), all the datasets achieve the best performance when the number of Gaussian mixture components reaches 5.

3) Hyperparameters λ_1 and λ_2: In the objective function, there are two hyperparameters (i.e., λ_1 and λ_2) corresponding to two modules. With an inappropriate value of λ_1, the SD module may be unable to maintain the data structure of the input HSIs during dimensionality reduction. In addition, the setting of λ_2 plays a significant role in the E module and a proper value can guarantee the data distribution to be well estimated. We set the hyperparameter λ_1 to 2, 3, 4, 5, and 6 and the hyperparameter λ_2 to 0.001, 0.01, 0.1, 1, and 10. According to the 3-D diagrams plotted in Fig. 4, the optimal values are selected as 3 and 0.1.

D. Detection Performance

By quantitatively comparing these detectors, Tables I and II list the AUC score of (TPR, FPR) and (FPR, τ) of the six popular methods and the proposed method on the four real HSIs. Meanwhile, to visually investigate the detection results and performance, Fig. 5 shows the visual detection maps of four HSIs obtained by the aforementioned algorithms.

First, by observing and analyzing the visual detection maps in Fig. 5, we notice that the proposed method can completely and markedly detect anomalies with different scales and

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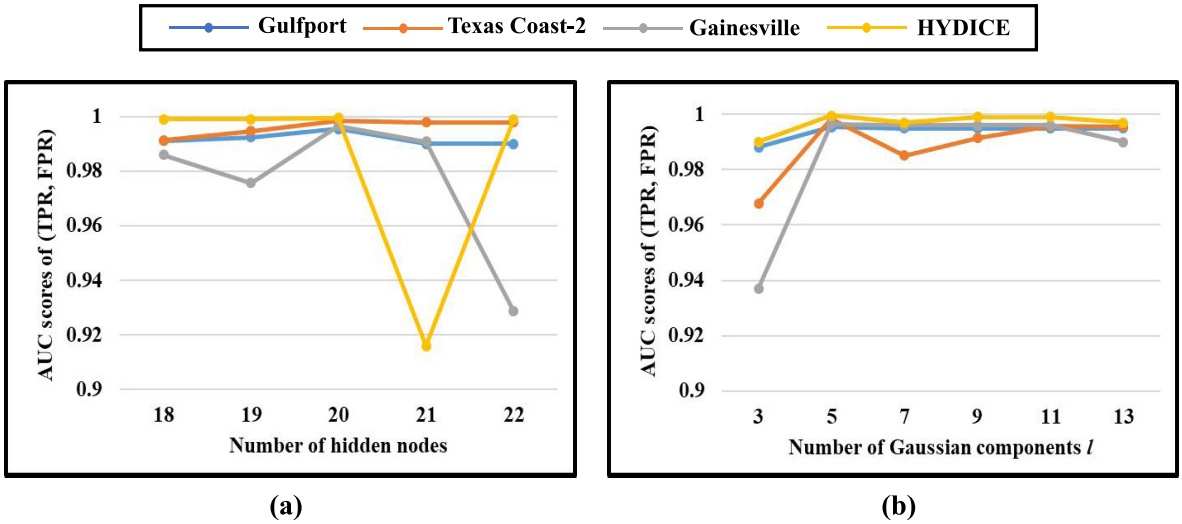


Fig. 3. Parameter analysis on different data sets. (a) Number of hidden nodes. (b) Number of Gaussian components.

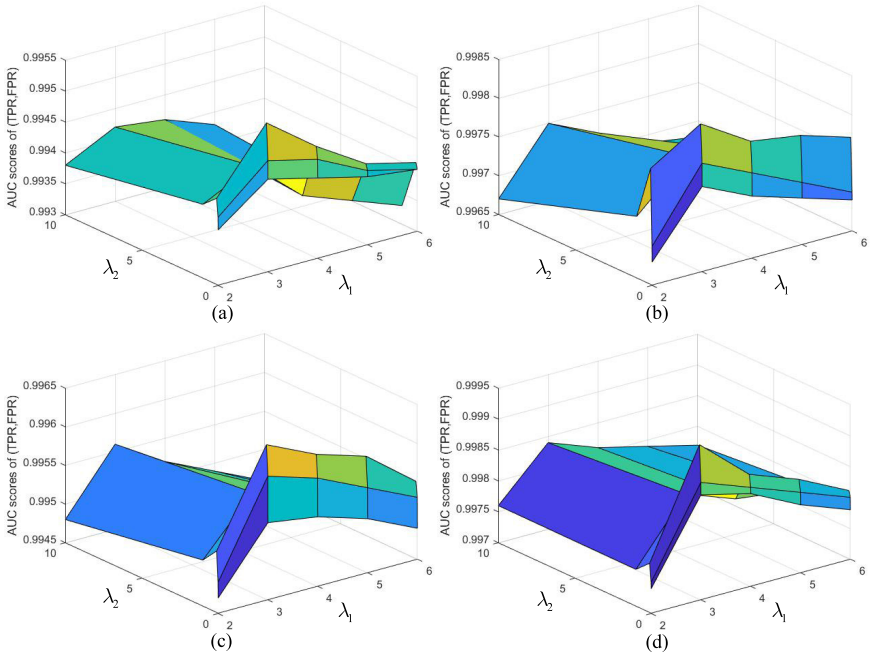


Fig. 4. Parameter analysis of the hyperparameters λ_1 and λ_2 on (a) Gulfport, (b) Texas Coast-2, (c) Gainesville, and (d) HYDICE.

positions, while other methods are more likely to lose the anomalies or can only show the vague targets. Besides, the proposed method can effectively suppress the background with rich redundant information, while the comparative algorithms are more vulnerable to background intrusion.

Specifically, for the Gulfport dataset, the proposed method can precisely locate all three airplanes with different sizes and fully highlight them. Although the LSMAD and PTA methods also show the same detection capability as the proposed method, each of them has its own disadvantages. The LSMAD fails to preserve the edge details of the small airplanes and the PTA shows a poor performance on suppressing the background. Noticeably, the RX and AAE methods completely miss all the anomalies. As for the LRASR and LSDM-MoG methods, they can only locate the largest targets, but the shapes of small ones are blurry.

For the Texas Coast-2 dataset, it is obvious that the proposed method can not only fully and accurately detect all the anomalies but also effectively suppress the background. On the contrary, the LSMAD method almost loses all the anomalies. In addition, the RX, LRASR, and LSDM-MoG methods miss anomalies to various degrees. As for the PTA method, it can find anomalies, but almost all the detected anomalies are mixed in the surrounding background. Furthermore, most anomalies detected by the AAE method are messy and unclear.

For the Gainesville dataset, it can be observed that the proposed method has a strong ability on extracting small anomalies that are disorganized and uneven distributed. Otherwise, none of the three methods, RX, LSMAD, and LSDM-MoG, can detect all the targets. Meanwhile, the AAE method misses almost all the anomalies. As for the LRASR

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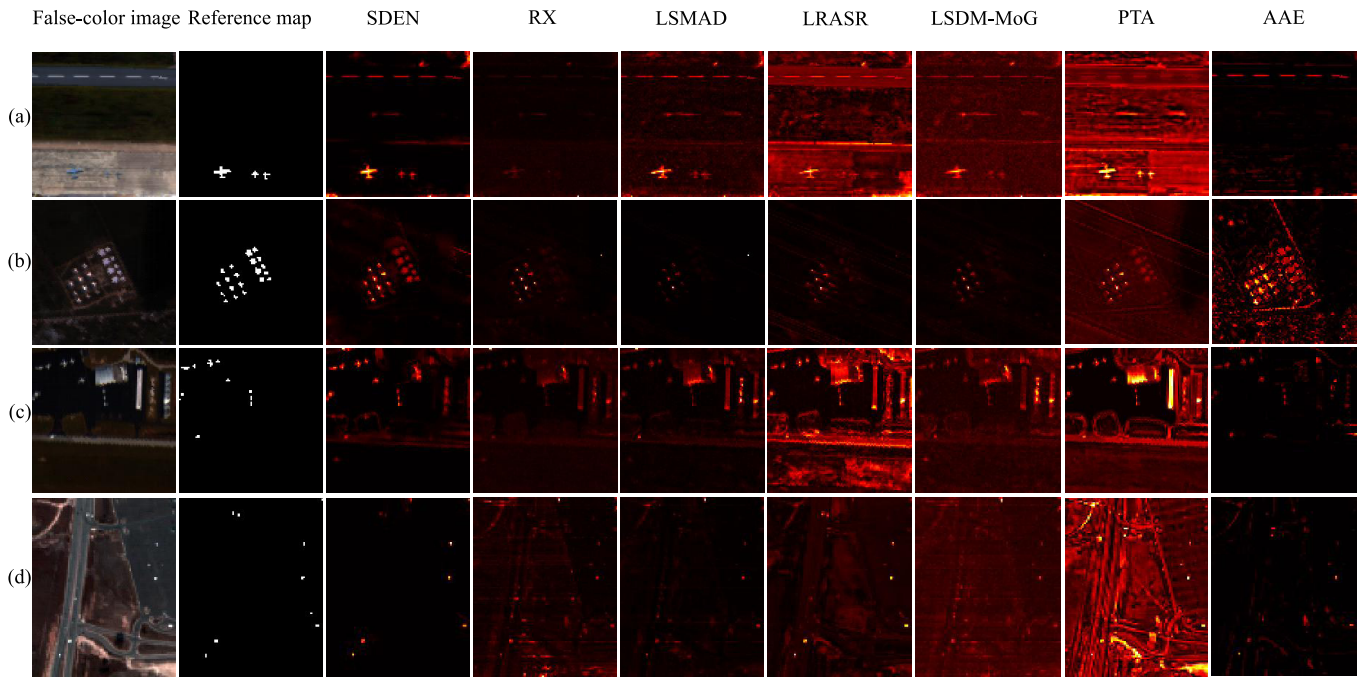


Fig. 5. False-color images, reference maps, and detection maps of the compared methods for (a) Gulfport, (b) Texas Coast-2, (c) Gainesville, and (d) HYDICE.

TABLE I
AUC SCORES OF (TPR, FPR) FOR THE COMPARED METHODS ON DIFFERENT DATASETS

Images	SDEN	RX	LSMAD	LRASR	LSDM -MoG	PTA	AAE
Gulfport	**0.9954**	0.9526	0.9869	0.9355	0.9591	0.9953	0.9443
Texas Coast-2	**0.9984**	0.9946	0.9964	0.9594	0.9847	0.9770	0.9681
Gainesville	**0.9965**	0.9513	0.9644	0.9546	0.9292	0.9385	0.9921
HYDICE	**0.9995**	0.9763	0.9881	0.9834	0.9537	0.9399	0.9218
Average	**0.9975**	0.9687	0.9840	0.9582	0.9567	0.9627	0.9566

TABLE II
AUC SCORES OF (FPR, τ) FOR THE COMPARED METHODS ON DIFFERENT DATASETS

Images	SDEN	RX	LSMAD	LRASR	LSDM -MoG	PTA	AAE
Gulfport	**0.0129**	0.0257	0.0552	0.1031	0.0998	0.1889	0.0286
Texas Coast-2	0.0156	0.0145	**0.0030**	0.0190	0.0163	0.0695	0.0433
Gainesville	0.0193	0.0360	0.0193	0.0924	0.0750	0.0979	**0.0084**
HYDICE	**0.0018**	0.0390	0.0173	0.0513	0.0855	0.1620	0.0106
Average	**0.0124**	0.0288	0.0237	0.0665	0.0692	0.1296	0.0227

and PTA methods, they are more likely to be confused by the background.

For the HYDICE dataset, it is apparent that the proposed method does a good job on detecting anomaly targets and suppressing the background. Instead, although the LSMAD and AAE methods can separate the anomalies pixels and background pixels effectively, they fail to detect more anomalies. The RX, LRASR, and LSDM-MoG methods mix anomalous targets with the surrounding background. Moreover, the anomalies of PTA method are more likely contaminated by the disturbing environment.

As shown in Tables I and II, the AUC scores of (TPR, FPR) and (FPR, τ) are consistent with the visual detection maps in Fig. 5. It is worth noting that the proposed method outperforms other methods with the highest AUC score of (TPR, FPR) on every HSI and the lowest average AUC score of (FPR, τ). Although the LSMAD algorithm also provides a good detection result with a high AUC score of (TPR, FPR), it fails to discriminate anomalies and the surrounding background. Besides, the LRASR, LSDM-MoG, and AAE algorithms cannot alleviate background and noise interference due to a high false alarm rate. As for the RX algorithm, it shows a robust detection performance on simple scenarios but is susceptible to the contamination of background. With the highest average AUC score of (FPR, τ), the PTA algorithm is weak in background-anomaly separability, which leads to a very poor detection performance.

Meanwhile, for clearly and intuitively demonstrating the detection performance among these methods and the proposed method, we plot the ROC curves of (TPR, FPR) and (FPR, τ) of seven approaches on the four HSIs. As shown in Figs. 6 and 7, the ROC curves of (TPR, FPR) of the proposed method lie nearer the top-left corner, and it is noticeable that the proposed method can obtain a high probability

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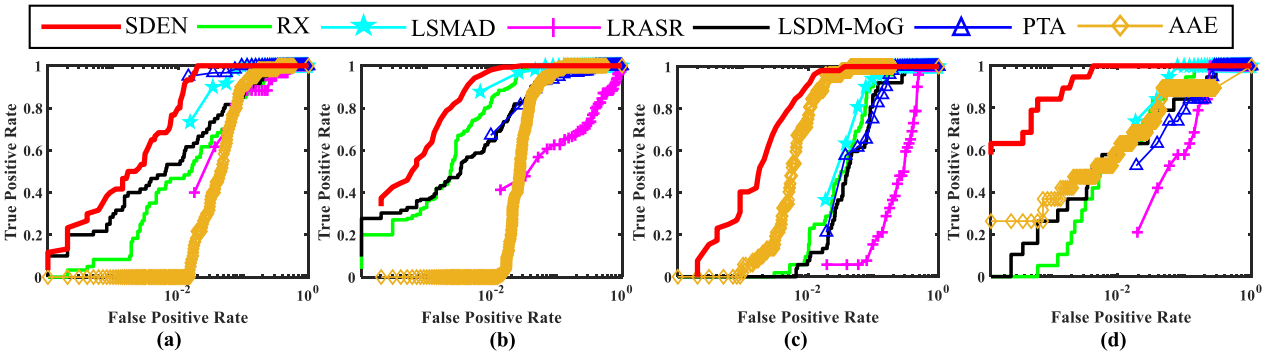


Fig. 6. ROC curves of (TPR, FPR) for the algorithms on (a) Gulfport, (b) Texas Coast-2, (c) Gainesville, and (d) HYDICE.

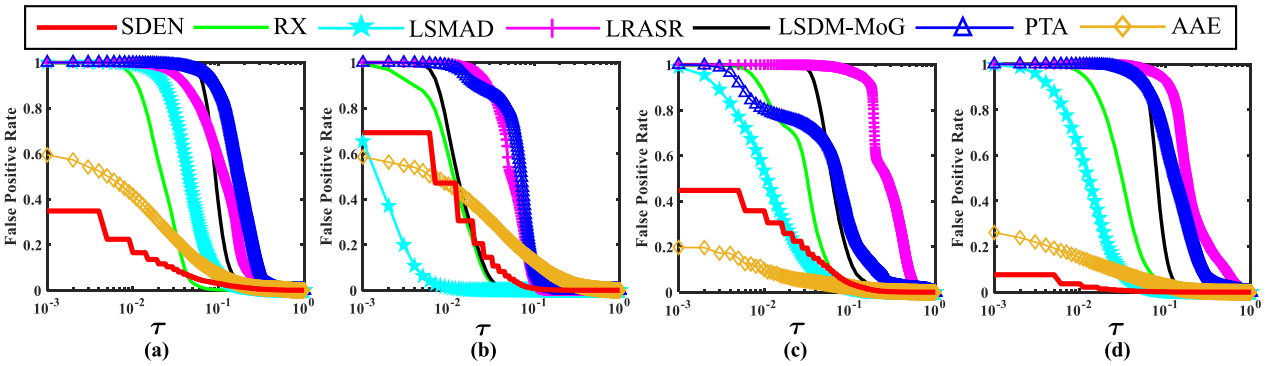


Fig. 7. ROC curves of (FPR, τ) for the algorithms on (a) Gulfport, (b) Texas Coast-2, (c) Gainesville, and (d) HYDICE.

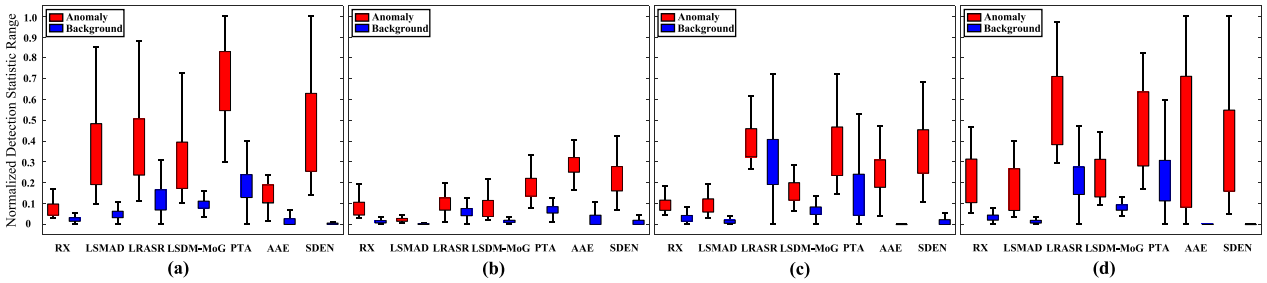


Fig. 8. Background-anomaly separation analysis of the compared methods on (a) Gulfport, (b) Texas Coast-2, (c) Gainesville, and (d) HYDICE.

of detection with a low false alarm and provide the most excellent detection performance. As for the ROC curves of (FPR, τ), although the LSMAD method has a strong ability to suppress the interference of background, it is unable to detect complete and accurate anomalies. On the contrary, with simultaneously discovering anomalous targets and suppressing surrounding background, the proposed method further shows its outstanding detection performance.

For further assessment, we employ the box-whisker plot to analyze the ability of separating the anomalies and the background and observe the effect of suppressing the background. As shown in Fig. 8, red boxes indicate anomalies and blue boxes represent the background. For the Gulfport dataset, an obvious gap is generated between the main anomaly box and background box, and the background box almost disappears, which demonstrates that the proposed method has an excellent ability of anomaly-background separation and can do a good job in background suppression. Other methods, including RX, LSMAD, LSDM-MoG, and AAE, cannot effectively distinguish anomaly and background. Meanwhile, the capabilities of suppressing the background for LRASR and PTA are very poor. For the Texas Coast-2 dataset, according to the box plot, it is apparent that the proposed method can separate to the great extent and effectively suppress the background. Although the AAE can also establish a clear gap, the capability of suppressing the background is very poor. As for the LRASR and PTA methods, the gap between the anomaly box and background box is totally unclear, which leads to poor performance. For the Gainesville dataset, it is obvious that the proposed method shows a noticeable gap compared to any other method, while the anomaly box is overlapped with the background box for LRASR, LSDM-MoG, and PTA. In addition, the blue boxes representing the background of LRASR and PTA are large, which illustrates the weakness of suppressing the background. For the HYDICE dataset, the proposed method does a good job in suppressing

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TABLE III
AUC SCORES OF (TPR, FPR) FOR ABLATION STUDY

Images	Est	AE+Est (two-step)	LIAE+Est (two-step)	SD+Est (two-step)	SDEN
Gulfport	0.9494	0.9899	0.9912	0.9923	**0.9954**
Texas Coast-2	0.9512	0.9613	0.9662	0.9765	**0.9984**
Gainesville	0.9448	0.9345	0.9521	0.9658	**0.9965**
HYDICE	0.9773	0.9948	0.9937	0.9944	**0.9995**
Average	0.9557	0.9701	0.9758	0.9823	**0.9975**

TABLE IV
AUC SCORES OF (FPR, τ) FOR ABLATION STUDY

Images	Est	AE+Est (two-step)	LIAE+Est (two-step)	SD+Est (two-step)	SDEN
Gulfport	0.0366	0.0271	0.0427	**0.0052**	0.0129
Texas Coast-2	0.0482	0.0278	0.0297	0.0298	**0.0156**
Gainesville	0.0607	0.0534	0.0532	0.0676	**0.0193**
HYDICE	0.0471	0.0560	0.0438	0.0745	**0.0018**
Average	0.0482	0.0411	0.0424	0.0443	**0.0124**

the background and the gap between anomaly box and background box is obvious. On the contrary, the LRASR and PTA methods are unable to effectively suppress the background. Overall, the proposed method can prominently reveal the capability of discriminating anomaly and background.

E. Ablation Study

For the sake of investigating the effect of each component in the proposed method, we conduct an ablation study in terms of AUC score of (TPR, FPR) and (FPR, τ) to assess the detection quality of five different methods. The first method applies only the estimation network Est, which is based on a GMM to directly detect anomalies on the input hyperspectral data. The second to fourth methods are all two-step approaches whose training procedures in the two steps are separated. As for the AE + Est (two-step), we first adopt an original AE to exploit the latent manifold embedded in the input data. Subsequently, the estimation network is employed to detect anomalies according to the latent features. As for the LIAE + Est (two-step), a local-invariant constraint is applied in the original AE to maintain the local connectivity and further perform the density estimation. Taking advantage of MCMC to preserve the data distribution, SD + Est (two-step) adopts the SD module to extract features and the latent features are used to detect anomalies. The last method SDEN acts as an end-to-end model and jointly optimizes the proposed SD module and E module to perform the dimensionality reduction and detect anomalies based on the density estimation. The detection results are shown in Tables III and IV. In terms of detection accuracy, it is obvious that the curse of dimensionality makes it difficult for the Est to perform density estimation. Otherwise, the two-step approaches achieve higher scores because of the feature extraction. Compared with the AE-Est (two-step), the LIAE-Est (two-step) obtains a higher accuracy due to the local-invariant constraint. In addition, an SD-aware module (i.e., SD module) is taken into consideration in the SD-Est (two-step), which leads to a better detection performance than the LIAE-Est (two-step). Moreover, it is apparent that the SDEN reaches the top compared with other approaches since it jointly optimizes the SD module and the E module. As for the false alarm rate, the SDEN almost obtains the lowest values on the four HSIs except for the Gulfport dataset and the value is only a little higher than the SD + Est (two-step), which further demonstrates the superiority of the proposed SDEN. In general, with the highest average AUC score of (TPR, FPR) and the lowest average AUC score of (FPR, τ), the SDEN shows a notable improvement and offers the best detection performance.

V. CONCLUSION

For deep unsupervised HAD, we have proposed the SDEN, including the SD module and the E module. Our key insight of the SD module is that the centroids in the raw input space remain to be centroids in the low-dimensional manifold. In particular, the local-invariant constraint facilitates the designed SD module to preserve the distribution of the raw input HSIs in a low-dimensional manifold, in which the E module distinguishes an anomaly according to the similarity and discrepancy between the anomaly and the background. With the learnable modules, the developed SDEN jointly learns feature extraction and anomaly detection in an end-to-end manner, thus being capable of detecting anomalies in the high-dimensional HSIs with complex contents in unsupervised settings.

Moreover, extensive and effective experiments ensure the reliability and accuracy of the proposed model. The quantitative evaluation experiments demonstrate that the proposed method is superior to those state-of-the-art methods in four datasets and the visual evaluation experiments show an excellent performance in HSIs. However, the random selection of the first centroid represents a limitation to the proposed model, which will be the focus of our future work.

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Weiying Xie (Member, IEEE) received the B.S. degree in electronic information science and technology from the University of Jinan, Jinan, China, in 2011, the M.S. degree in communication and information systems from Lanzhou University, Lanzhou, China, in 2014, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2017.

She is an Associate Professor with the State Key Laboratory of Integrated Services Networks, Xidian University. She has published more than 30 articles in refereed journals and conferences, including IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON CYBERNETICS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, AAAI Conference on Artificial Intelligence, and *Pattern Recognition*. Her research interests include neural networks, machine learning, hyperspectral image processing, and high-performance computing.

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Shuran Fan received the B.E. degree in communication engineering from the Zhejiang University of Technology, Hangzhou, China, in 2020. She is pursuing the M.S. degree with the State Key Laboratory of Integrated Services Networks, Image Coding and Processing Center, Xidian University, Xi'an, China.

Her research interests include hyperspectral images, anomaly detection, and deep learning.



Jiahui Qu received the B.S. degree in communication engineering from Yantai University, Yantai, China, in 2014, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in June 2020.

She was an exchange Ph.D. student with Mississippi State University, Starkville, MS, USA, from 2018 to 2019. She is a Lecturer with the State Key Laboratory of Integrated Services Networks, Xidian University. Her research interests include hyperspectral image detection, image fusion, neural networks, and deep learning.



Xianyun Wu was born in 1985. He received the Ph.D. degree from Xidian University, Xi'an, China, in 2012.

He is an Associate Professor at Xidian University. His research interests include remote sensing, image compression and processing, and high-performance computing.



Yanli Lu received the M.S. degree in computer application technology and the Ph.D. degree in signal and information processing from Air Force Engineering University, Xi'an, China, in 2005 and 2008, respectively.

She is an Associate Professor with the College of Air and Missile Defense, Air Force Engineering University. Her research interests focus on uncertain reasoning, intuitionistic fuzzy sets, rough sets, and evidence theory.



Qian Du (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland at Baltimore, Baltimore, MD, USA, in 2000.

She is the Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. She is also an Adjunct Professor with the College of Surveying and Geo-Informatics, Tongji University, Shanghai, China. Her research interests include hyperspectral remote sensing image analysis and applications, pattern classification, data compression, and neural networks.

Dr. Du is a fellow of the SPIE-International Society for Optics and Photonics. She received the 2010 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society. She was the Co-Chair of the Data Fusion Technical Committee of the IEEE Geoscience and Remote Sensing Society from 2009 to 2013 and the Chair of the Remote Sensing and Mapping Technical Committee of the International Association for Pattern Recognition from 2010 to 2014. She was the General Chair of the Fourth IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing held at Shanghai, in 2012. She has served as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the *Journal of Applied Remote Sensing*, and the IEEE SIGNAL PROCESSING LETTERS. Since 2016, she has been the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

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